

CampusConnect

Campus Opportunity Management Platform

Discover • Save • Track • Apply

Professional Project Presentation

Project Overview

- **What is CampusConnect?**

A centralized digital platform that helps students discover and manage internships, events, scholarships, clubs and other campus opportunities.

- **Technology Stack**

Frontend: React / HTML / CSS / JavaScript

Backend: Node.js + Express

API: RESTful services

Database: MySQL / PostgreSQL

Problem & Solution

- **The Challenge**

Students often depend on scattered notices, portals, messaging groups and department announcements. This makes opportunities difficult to discover and applications difficult to track.

- **The Solution**

CampusConnect creates one organized platform with search, filtering, bookmarking, application tracking and centralized opportunity information.

Core Features

- **01 — Search** Quickly discover relevant opportunities.
- **02 — Save** Bookmark opportunities for future review.
- **03 — Track** Monitor application progress and status.
- **04 — Apply** Access requirements, deadlines and application workflows.

Objectives & Scope

- **MVP Objective**

Deliver a reliable minimum viable platform focused on the complete student opportunity journey.

- **Priority Scope**

Authentication • Opportunity discovery • Search & filters • Save/bookmark • Application tracking • Basic administration

- **Success Principle**

Simple for students, manageable for administrators, and ready to scale.

User Stories

- 01. As a student, I want to create an account so I can manage my opportunities.
- 02. As a student, I want to search opportunities so I can find relevant options quickly.
- 03. As a student, I want to filter results so I can narrow down choices.
- 04. As a student, I want to save opportunities so I can revisit them later.
- 05. As a student, I want to track applications so I can monitor their status.
- 06. As a student, I want to view deadlines so I can apply on time.
- 07. As an admin, I want to manage listings so students receive accurate information.

Technical Architecture

- **Presentation Layer**

Responsive web interface • Student dashboard • Admin dashboard

- **Application Layer**

Authentication • Search • Opportunity management • Saved items • Application workflow

- **API Layer**

REST controllers • Validation • Authorization • JSON responses

- **Data Layer**

Relational database • Users • Opportunities • Saved records • Applications

REST API Endpoints

- POST /api/auth/register — Register a user
- POST /api/auth/login — Authenticate a user
- GET /api/opportunities — List, search and filter opportunities
- GET /api/opportunities/:id — Retrieve opportunity details
- POST /api/saved — Save an opportunity
- DELETE /api/saved/:id — Remove a saved opportunity
- POST /api/applications — Create an application record
- GET /api/applications — Retrieve application tracking data

Database Schema

- **Users** — user_id, name, email, password_hash, role
- **Opportunities** — opportunity_id, title, type, description, deadline, organization
- **Saved** — saved_id, user_id, opportunity_id, created_at
- **Applications** — application_id, user_id, opportunity_id, status, applied_at
- **Relationships** — User 1:N Saved • Opportunity 1:N Saved • User 1:N Applications • Opportunity 1:N Applications

Security & Requirements

- **Security Best Practices**

HTTPS/TLS • Secure password hashing • Authentication • Role-based authorization • Input validation • Parameterized queries • Secure token/session handling

- **Quality Requirements**

Performance • Availability • Usability • Maintainability • Scalability • Data integrity

Implementation Plan

- **Phase 01 — Discover** Requirements, personas, workflows and wireframes
- **Phase 02 — Design** UI system, database schema and API contracts
- **Phase 03 — Build** Frontend, backend, REST API and database integration
- **Phase 04 — Validate** Functional testing, security review and performance testing
- **Phase 05 — Launch** Deployment, monitoring, documentation and feedback

Future Enhancements

- Personalized opportunity recommendations
- Email and push notifications for deadlines
- Resume/profile-based opportunity matching
- Advanced analytics for administrators
- Dedicated mobile application
- Integration with college ERP, placement and learning systems

Ready to Build

- **CampusConnect** turns fragmented campus opportunities into one connected experience.
- **Discover. Save. Track. Apply.**
- Next step: convert the MVP blueprint into a working prototype.